Representation of Data

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Stem-and-leaf diagrams**

The data set $23, 25, 31, 34, 34, 38, 42, 45$ is to be shown as a stem-and-leaf diagram with stems $2, 3, 4$. Write out the diagram, and use it to find the median.

Stem-and-leaf: $2 \mid 3\,5$; $3 \mid 1\,4\,4\,8$; $4 \mid 2\,5$ (key: $2 \mid 3 = 23$). With 8 values, the median is the average of the 4th and 5th values: 34 and 34, so median $= 34$.

2 **[Foundational]** • **Target: Stem-and-leaf diagrams**

The data set $17, 19, 22, 26, 26, 29, 33, 37, 41$ is to be shown as a stem-and-leaf diagram with stems $1, 2, 3, 4$. Write out the diagram, and use it to find the range.

Stem-and-leaf: $1 \mid 7\,9$; $2 \mid 2\,6\,6\,9$; $3 \mid 3\,7$; $4 \mid 1$ (key: $1 \mid 7 = 17$). Range $= 41 - 17 = 24$.

3 **[Foundational]** • **Target: Histograms and frequency density**

A class interval of width 10 contains 25 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

Frequency density $= \frac{\text{frequency}}{\text{class width}} = \frac{25}{10} = 2.5$.

4 **[Foundational]** • **Target: Histograms and frequency density**

A class interval of width 4 contains 18 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

Frequency density $= \frac{\text{frequency}}{\text{class width}} = \frac{18}{4} = 4.5$.

5 **[Foundational]** • **Target: Histograms and frequency density**

A class interval of width 5 contains 30 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

Frequency density $= \frac{\text{frequency}}{\text{class width}} = \frac{30}{5} = 6$.

6 **[Foundational]** • **Target: Reading grouped frequency data from a bar chart**

The bar chart shows the number of books read by students in a reading challenge, grouped by number of books. Find the total number of students, and the modal group.

Total students $= 12 + 28 + 15 + 5 = 60$. The modal group (highest bar) is $3{-}5$ books, with 28 students.

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7 [Foundational] • Target: Reading grouped frequency data from a bar chart

Using the same reading-challenge data (0-2: 12, 3-5: 28, 6-8: 15, 9-11: 5 students), find what percent of students read 6 or more books.

Students reading 6 or more books: $15 + 5 = 20$. Percent of the total 60 students: $\frac{20}{60} \times 100\% \approx 33.3\%$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Measures of central tendency for grouped data

A grouped data set has class midpoints and frequencies: 5 (freq 4), 15 (freq 8), 25 (freq 6), 35 (freq 2). Estimate the mean.

$\bar{x} = \frac{5(4) + 15(8) + 25(6) + 35(2)}{4 + 8 + 6 + 2} = \frac{20 + 120 + 150 + 70}{20} = \frac{360}{20} = 18.$

9 [Intermediate] • Target: Measures of central tendency for grouped data

A grouped data set has class midpoints and frequencies: 10 (freq 3), 20 (freq 9), 30 (freq 5), 40 (freq 3). Estimate the mean.

$\bar{x} = \frac{10(3) + 20(9) + 30(5) + 40(3)}{3 + 9 + 5 + 3} = \frac{30 + 180 + 150 + 120}{20} = \frac{480}{20} = 24.$

10 [Intermediate] • Target: Box-and-whisker plots

A data set has minimum $= 8$, $Q_1 = 15$, median $= 22$, $Q_3 = 30$, and maximum $= 40$. State the interquartile range, and determine whether a value of 45 would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.

$IQR = Q_3 - Q_1 = 30 - 15 = 15$. Upper outlier boundary: $Q_3 + 1.5(IQR) = 30 + 22.5 = 52.5$. Since $45 < 52.5$, the value 45 would NOT be considered an outlier.

11 [Intermediate] • Target: Box-and-whisker plots

A data set has minimum $= 5$, $Q_1 = 12$, median $= 18$, $Q_3 = 27$, and maximum $= 50$. State the interquartile range, and determine whether the maximum value would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.

$IQR = Q_3 - Q_1 = 27 - 12 = 15$. Upper outlier boundary: $Q_3 + 1.5(IQR) = 27 + 22.5 = 49.5$. Since the maximum value $50 > 49.5$, it WOULD be considered an outlier.

12 [Intermediate] • Target: Box-and-whisker plots

A data set has $Q_1 = 20$ and $Q_3 = 32$. Find the lower outlier boundary, defined as any value below $Q_1 - 1.5 \times IQR$.

$IQR = Q_3 - Q_1 = 32 - 20 = 12$. $1.5 \times IQR = 1.5 \times 12 = 18$. Lower outlier boundary $= 20 - 18 = 2$.

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13 [Intermediate] • Target: Skewness from summary statistics

A data set has mean $= 42$, median $= 38$, and mode $= 35$. Describe the skewness of the distribution.

Since mean $>$ median $>$ mode, the distribution is positively skewed (skewed to the right), with a longer tail of higher values pulling the mean above the median.

14 [Intermediate] • Target: Skewness from summary statistics

A data set has mean $= 28$, median $= 32$, and mode $= 35$. Describe the skewness of the distribution.

Since mean $<$ median $<$ mode, the distribution is negatively skewed (skewed to the left), with a longer tail of lower values pulling the mean below the median.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 11.01 Kinematics

Two classes take the same test. Class A has mean 68 and standard deviation 5. Class B has mean 71 and standard deviation 12. Compare the two classes' performance and consistency.

Class B has a higher mean score (71 vs. 68), so on average Class B performed better. However, Class A has a much smaller standard deviation (5 vs. 12), meaning Class A's scores were more consistent/less spread out than Class B's.

16 [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 12.01 Forces & Equilibrium

Two machines fill bottles with a target volume of 500 mL. Machine X has mean fill volume 500 mL and standard deviation 2 mL. Machine Y has mean fill volume 498 mL and standard deviation 8 mL. Compare the two machines' accuracy and consistency.

Machine X's mean (500 mL) matches the target exactly, while Machine Y's mean (498 mL) is 2 mL below target. Machine X also has a much smaller standard deviation (2 mL vs. 8 mL), meaning its fill volumes are far more consistent. Machine X is therefore both more accurate and more consistent than Machine Y.

17 [Challenging] • Target: A well-designed word problem: analyzing delivery times • Cross-Lesson: 13.01 Newton's Laws

This question has two parts. A delivery company records the delivery times (in minutes) for 50 packages: mean $= 34$ minutes, standard deviation $= 6$ minutes. Company policy states any delivery beyond 2 standard deviations above the mean is classified as late. (a) Find the delivery-time threshold, in minutes, above which a delivery is classified as late. (b) If a customer's package took 44 minutes to deliver, determine, using the standard deviation, whether this delivery should be classified as late, and state how many standard deviations from the mean this value lies.

(a) Threshold $= 34 + 2(6) = 34 + 12 = 46$ minutes. (b) 44 minutes is $\frac{44 - 34}{6} = \frac{10}{6} \approx 1.67$ standard deviations above the mean. Since $1.67 < 2$, this delivery is NOT classified as late.

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18 [Challenging] • Target: Skewness from summary statistics • Cross-Lesson: 14.01 Energy, Work & Power

A grouped frequency table has class intervals of width 10 starting at 9, with frequencies $[6, 6, 14, 11, 12]$ respectively. Using midpoints, find the mean.

Final Answer: 1836/49. Midpoints: $[14, 24, 34, 44, 54]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1836}{49} = 1836/49$

19 [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 16.01 Permutations & Combinations

A grouped frequency table has class intervals of width 8 starting at 11, with frequencies $[5, 10, 9, 11, 14]$ respectively. Using midpoints, find the mean.

Final Answer: 1671/49. Midpoints: $[15, 23, 31, 39, 47]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1671}{49} = 1671/49$

20 [Challenging] • Target: A well-designed word problem: analyzing delivery times • Cross-Lesson: 17.01 Discrete Random Variables

A grouped frequency table has class intervals of width 10 starting at 15, with frequencies $[13, 12, 13, 4, 4]$ respectively. Using midpoints, find the mean.

Final Answer: 790/23. Midpoints: $[20, 30, 40, 50, 60]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1580}{46} = 790/23$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Stem-and-Leaf Diagrams & Box Plots

In an ordered stem-and-leaf diagram of 15 test scores, the key states $3 \mid 5 = 35$. The leaves for stem 4 are $1, 3, 5, 8, 8$. What is the median score if there are 4 scores below stem 4?

a. 45 ✓

b. 43

c. 48

d. 41

For $n = 15$, the median is the 8th value: 4 values are in stem 3, so in stem 4 we count the 5th, 6th, 7th, 8th values (1, 3, 5, 8). The 8th value is 45.

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22 [Foundational] • Target: Stem-and-Leaf Diagrams & Box Plots

A data set has minimum $= 12$, $Q_1 = 20$, median $= 28$, $Q_3 = 36$, and maximum $= 52$. What is the interquartile range (IQR)?

a. 16 ✓

b. 24

c. 40

d. 8

$\text{IQR} = Q_3 - Q_1 = 36 - 20 = 16$.

23 [Foundational] • Target: Stem-and-Leaf Diagrams & Box Plots

Using the $1.5 \times \text{IQR}$ rule on the data from the previous question, determine if the maximum value 52 is an outlier.

a. No, the upper outlier boundary is 60 ✓

b. Yes, any value above 48 is an outlier

c. Yes, any value above 50 is an outlier

d. No, because the range is less than 50

Boundary $= Q_3 + 1.5 \times \text{IQR} = 36 + 1.5(16) = 36 + 24 = 60$. Since $52 \leq 60$, it is not an outlier.

24 [Foundational] • Target: Histograms & Frequency Density

In a histogram, the class interval $20 \leq x < 35$ has a frequency of 45. What is the frequency density for this bar?

a. 3.0 ✓

b. 1.5

c. 4.5

d. 2.25

Class width $= 35 - 20 = 15$. Frequency density $= \frac{\text{Frequency}}{\text{Class width}} = \frac{45}{15} = 3.0$.

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25 [Foundational] • Target: Histograms & Frequency Density

On a histogram, a bar representing the interval $10 \leq x < 20$ has a width of 2 cm and a height of 5 cm, representing 30 items. What area represents 1 item?

a. $\frac{1}{3}$ cm^2 ✓

b. $\frac{1}{2}$ cm^2

c. 3 cm^2

d. 1 cm^2

Area of bar $= 2 \times 5 = 10$ cm^2. Since this represents 30 items, each item is represented by $10/30 = 1/3$ cm^2.

26 [Foundational] • Target: Histograms & Frequency Density

Why is frequency density, rather than raw frequency, plotted on the vertical axis of a histogram for grouped continuous data?

a. **Because class widths may be unequal and area must be proportional to frequency** ✓

b. Because continuous data cannot be plotted with integer frequencies

c. To ensure the total height of all bars equals 100

d. To eliminate the need for class boundaries

When classes have unequal widths, plotting frequency density ensures that the area of each rectangle is directly proportional to the frequency of that class.

27 [Foundational] • Target: Cumulative Frequency Curves & Percentiles

When plotting a cumulative frequency graph for grouped continuous data, at which points should the cumulative frequencies be plotted?

a. **At the upper class boundaries** ✓

b. At the midpoints of each class

c. At the lower class boundaries

d. At the class frequencies

Cumulative frequency represents the total number of observations up to that value, so it must be plotted at the upper boundary of each class interval.

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28 [Foundational] • Target: Cumulative Frequency Curves & Percentiles

From a cumulative frequency curve of 200 students, the 80th percentile corresponds to which cumulative frequency?

a. **160** ✓

b. 80

c. 120

d. 180

The 80th percentile corresponds to $0.80 \times 200 = 160$ on the vertical cumulative frequency axis.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Cumulative Frequency Curves & Percentiles

Using linear interpolation on grouped data, the median lies in the class $40 \leq x < 50$ with frequency 25 and cumulative frequency before this class equal to 35. If total $n = 100$, find the estimated median.

a. 46.0 ✓

b. 45.0

c. 44.5

d. 47.2

Median position $= 100/2 = 50$. We need $50 - 35 = 15$ more items out of 25. Median $= 40 + \frac{15}{25}(10) = 40 + 6 = 46.0$.

30 [Intermediate] • Target: Linear Coding & Transformations of Summary Stats

A data set x has mean $\bar{x} = 50$ and variance $s_x^2 = 16$. A new variable is defined as $y = 3x - 10$. Find the mean $\bar{y}$ and standard deviation s_y.

a. $\bar{y} = 140$, $s_y = 12$ ✓

b. $\bar{y} = 140$, $s_y = 36$

c. $\bar{y} = 150$, $s_y = 12$

d. $\bar{y} = 140$, $s_y = 4$

Mean: $\bar{y} = 3(50) - 10 = 140$. Standard deviation: $s_y = |a|s_x = 3(\sqrt{16}) = 3(4) = 12$.

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31 [Intermediate] • Target: Linear Coding & Transformations of Summary Stats

A variable t is coded as $u = \frac{t-200}{10}$. For 50 observations, $\sum u = 85$ and $\sum u^2 = 320$. Find the actual mean of t.

a. 217 ✓

b. 215

c. 208.5

d. 220

Mean of u: $\bar{u} = 85/50 = 1.7$. Actual mean $\bar{t} = 10\bar{u} + 200 = 10(1.7) + 200 = 217$.

32 [Intermediate] • Target: Linear Coding & Transformations of Summary Stats

For the data in the previous question, calculate the variance of the original variable t.

a. 351 ✓

b. 35.1

c. 320

d. 500

$\text{Var}(u) = \frac{\sum u^2}{n} - \bar{u}^2 = \frac{320}{50} - (1.7)^2 = 6.4 - 2.89 = 3.51$. Then $\text{Var}(t) = 10^2 \times \text{Var}(u) = 100 \times 3.51 = 351$.

33 [Intermediate] • Target: Stem-and-leaf diagrams

A grouped frequency table has class intervals of width 10 starting at 17, with frequencies $[5, 13, 9, 12, 3]$ respectively. Using midpoints, find the mean.

a. 899/21

b. 836/21

c. **857/21** ✓

d. 878/21

Midpoints: $[22, 32, 42, 52, 62]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1714}{42} = 857/21$

34 [Intermediate] • Target: Histograms and frequency density

A grouped frequency table has class intervals of width 8 starting at 9, with frequencies $[15, 9, 11, 4, 5]$ respectively. Using midpoints, find the mean.

a. **269/11** ✓

b. 280/11

c. 258/11

d. 291/11

Midpoints: $[13, 21, 29, 37, 45]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1076}{44} = 269/11$

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35 [Intermediate] • Target: Reading grouped frequency data from a bar chart

A grouped frequency table has class intervals of width 8 starting at 19, with frequencies $[9, 9, 13, 13, 13]$ respectively. Using midpoints, find the mean.

a. 792/19

b. 754/19

c. 811/19

d. **773/19** ✓

Midpoints: $[23, 31, 39, 47, 55]$. Mean $= \frac{\sum fx}{\sum f} = \frac{2319}{57} = 773/19$

36 [Intermediate] • Target: Measures of central tendency for grouped data

A grouped frequency table has class intervals of width 8 starting at 17, with frequencies $[3, 15, 12, 8, 12]$ respectively. Using midpoints, find the mean.

a. 944/25

b. 1019/25

c. 994/25

d. **969/25** ✓

Midpoints: $[21, 29, 37, 45, 53]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1938}{50} = 969/25$

37 [Intermediate] • Target: Box-and-whisker plots

A grouped frequency table has class intervals of width 8 starting at 11, with frequencies $[14, 9, 12, 7, 10]$ respectively. Using midpoints, find the mean.

a. 370/13

b. 396/13

c. **383/13** ✓

d. 409/13

Midpoints: $[15, 23, 31, 39, 47]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1532}{52} = 383/13$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Skewness from summary statistics • Cross-Lesson: 14.01 Energy, Work & Power

A grouped frequency table has class intervals of width 8 starting at 9, with frequencies $[11, 7, 7, 15, 10]$ respectively. Using midpoints, find the mean.

a. 799/25

b. 774/25

c. **749/25** ✓

d. 724/25

Midpoints: $[13, 21, 29, 37, 45]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1498}{50} = 749/25$

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39 [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 16.01 Permutations & Combinations

A grouped frequency table has class intervals of width 8 starting at 6, with frequencies $[6, 6, 11, 7, 10]$ respectively. Using midpoints, find the mean.

a. 144/5

b. 134/5

c. **139/5 ✓**

d. 149/5

Midpoints: $[10, 18, 26, 34, 42]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1112}{40} = 139/5$

40 [Challenging] • Target: A well-designed word problem: analyzing delivery times • Cross-Lesson: 17.01 Discrete Random Variables

A grouped frequency table has class intervals of width 8 starting at 16, with frequencies $[15, 3, 5, 15, 2]$ respectively. Using midpoints, find the mean.

a. 161/5

b. 171/5

c. 176/5

d. **166/5 ✓**

Midpoints: $[20, 28, 36, 44, 52]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1328}{40} = 166/5$

41 [Challenging] • Target: Stem-and-Leaf Diagrams & Box Plots • Cross-Lesson: 18.01 Binomial Distribution

A grouped frequency table has class intervals of width 10 starting at 1, with frequencies $[6, 13, 5, 13, 13]$ respectively. Using midpoints, find the mean.

a. **144/5 ✓**

b. 154/5

c. 149/5

d. 139/5

Midpoints: $[6, 16, 26, 36, 46]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1440}{50} = 144/5$

42 [Challenging] • Target: Histograms & Frequency Density • Cross-Lesson: 19.01 Normal Distribution

A grouped frequency table has class intervals of width 10 starting at 8, with frequencies $[9, 7, 9, 2, 3]$ respectively. Using midpoints, find the mean.

a. 79/3

b. 88/3

c. **82/3 ✓**

d. 85/3

Midpoints: $[13, 23, 33, 43, 53]$. Mean $= \frac{\sum fx}{\sum f} = \frac{820}{30} = 82/3$

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43 [Challenging] • Target: Cumulative Frequency Curves & Percentiles • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A grouped frequency table has class intervals of width 10 starting at 9, with frequencies $[6, 7, 13, 7]$ respectively. Using midpoints, find the mean.

a. 323/11

b. **334/11** ✓

c. 345/11

d. 356/11

Midpoints: $[14, 24, 34, 44]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1002}{33} = 334/11$

44 [Challenging] • Target: Linear Coding & Transformations of Summary Stats • Cross-Lesson: 3.01 Graphs & Transformations

A grouped frequency table has class intervals of width 10 starting at 10, with frequencies $[15, 6, 6, 6, 7]$ respectively. Using midpoints, find the mean.

a. 32

b. 33

c. 30

d. **31** ✓

Midpoints: $[15, 25, 35, 45, 55]$. Mean $= \frac{\sum fx}{\sum f} = \frac{1240}{40} = 31$

45 [Challenging] • Target: Stem-and-leaf diagrams • Cross-Lesson: 4.01 Coordinate Geometry

A grouped frequency table has class intervals of width 10 starting at 11, with frequencies $[14, 13, 4, 3]$ respectively. Using midpoints, find the mean.

a. 405/17

b. **422/17** ✓

c. 456/17

d. 439/17

Midpoints: $[16, 26, 36, 46]$. Mean $= \frac{\sum fx}{\sum f} = \frac{844}{34} = 422/17$

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